

B E T S

Institute of Technology

Research on

CYBERSECURITY DOMAINS

A Comprehensive Overview of Key Domains, Skills, Tools & Career Paths

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Cybersecurity is a rapidly evolving field comprising numerous specialized domains, each addressing distinct aspects of digital security. This report examines ten critical cybersecurity domains, providing an in-depth analysis of their definitions, key responsibilities, required technical and soft skills, commonly used tools, relevant industry certifications, and career progression opportunities. Understanding these domains is essential for aspiring professionals seeking to build a successful career in cybersecurity.

1. Penetration Testing

Definition & Overview: Penetration testing involves simulating real-world cyberattacks against systems, networks, and applications to identify exploitable vulnerabilities before malicious actors can leverage them. Testers follow structured methodologies such as OWASP and PTES to assess security posture.

Key Responsibilities: Conducting authorized attacks on infrastructure, writing detailed vulnerability reports, recommending remediation strategies, and retesting after fixes are applied.

Technical Skills: Kali Linux, Metasploit, Burp Suite, Nmap, Wireshark, Python/Bash scripting, exploit development.

Soft Skills: Analytical thinking, persistence, strong written communication, ethical judgment, attention to detail.

Tools Used: Metasploit Framework, Burp Suite Professional, Nmap, Cobalt Strike, Nessus, OWASP ZAP.

Certifications: OSCP (Offensive Security), CEH (EC-Council), GPEN (GIAC), CompTIA PenTest+.

Career Path: Junior Penetration Tester → Senior Penetration Tester → Lead Consultant → Red Team Manager → Chief Security Officer.

2. Digital Forensics

Definition & Overview: Digital forensics is the practice of collecting, preserving, analyzing, and presenting digital evidence from computers, mobile devices, and networks. It plays a critical role in criminal investigations, incident response, and legal proceedings.

Key Responsibilities: Acquiring forensic images, preserving chain of custody, analyzing file systems and memory dumps, recovering deleted data, and preparing court-admissible reports.

Technical Skills: Disk imaging, memory forensics, file system analysis, log correlation, malware triage, mobile device extraction.

Soft Skills: Meticulous attention to detail, patience, objectivity, excellent documentation skills, courtroom presentation ability.

Tools Used: EnCase, FTK (Forensic Toolkit), Autopsy, Volatility, Cellebrite, X-Ways Forensics.

Certifications: CHFI (EC-Council), GCFE (GIAC), EnCE (Guidance Software), ACE (AccessData).

Career Path: Forensic Analyst → Senior Forensic Examiner → Incident Response Lead → Digital Forensics Manager → CISO.

3. SOC Analyst

Definition & Overview: A Security Operations Center (SOC) Analyst monitors an organization's IT environment around the clock, detecting, analyzing, and responding to security incidents in real time. SOC Analysts serve as the frontline defenders against cyber threats.

Key Responsibilities: Monitoring SIEM dashboards, triaging security alerts, escalating confirmed incidents, conducting initial threat analysis, and documenting incident timelines.

Technical Skills: SIEM administration, log analysis, packet capture analysis, endpoint detection and response (EDR), threat hunting basics.

Soft Skills: Situational awareness, ability to work under pressure, strong communication, teamwork, critical thinking.

Tools Used: Splunk, IBM QRadar, Microsoft Sentinel, CrowdStrike Falcon, Elastic SIEM, TheHive.

Certifications: CompTIA Security+, CompTIA CySA+, Splunk Certified User, SC-200 (Microsoft), BTL1 (Blue Team Level 1).

Career Path: SOC Analyst Tier 1 → Tier 2 → Tier 3 (Threat Hunter) → SOC Manager → Director of Security Operations.

4. Malware Analysis

Definition & Overview: Malware analysis involves dissecting malicious software to understand its behavior, origin, propagation methods, and impact. Analysts use static and dynamic techniques to reverse-engineer malware samples and develop detection signatures.

Key Responsibilities: Performing static and dynamic analysis of malware samples, reverse-engineering binaries, writing YARA rules and detection signatures, and producing intelligence reports.

Technical Skills: Assembly language, reverse engineering, sandboxing, behavioral analysis, memory forensics, programming in C/C++ and Python.

Soft Skills: Curiosity, persistence, analytical reasoning, technical writing, pattern recognition.

Tools Used: IDA Pro, Ghidra, x64dbg, Cuckoo Sandbox, YARA, PEStudio, Any.Run.

Certifications: GREM (GIAC), CEH (EC-Council), OSCP (Offensive Security), CREA (Certified Reverse Engineering Analyst).

Career Path: Junior Malware Analyst → Senior Malware Analyst → Threat Researcher → Malware Lab Director.

5. Threat Intelligence

Definition & Overview: Threat intelligence focuses on collecting, processing, and analyzing data about current and emerging cyber threats. The goal is to provide actionable insights that enable proactive defense and informed decision-making across the organization.

Key Responsibilities: Gathering intelligence from open-source and proprietary feeds, mapping adversary tactics to the MITRE ATT&CK framework, producing threat briefs, and supporting incident response with contextual data.

Technical Skills: OSINT collection, indicator of compromise (IoC) analysis, MITRE ATT&CK mapping, threat modeling, STIX/TAXII protocols.

Soft Skills: Critical thinking, research skills, clear communication, geopolitical awareness, collaboration.

Tools Used: MISP, Recorded Future, Anomali ThreatStream, VirusTotal, Shodan, Maltego.

Certifications: CTIA (EC-Council), GCTI (GIAC), CompTIA CySA+, SANS FOR578.

Career Path: Threat Intelligence Analyst → Senior Analyst → Threat Intel Lead → Director of Cyber Intelligence.

6. Cloud Security

Definition & Overview: Cloud security involves protecting data, applications, and infrastructure hosted on cloud platforms such as AWS, Azure, and Google Cloud. It addresses unique challenges including shared responsibility models, identity management, and multi-tenant architectures.

Key Responsibilities: Designing secure cloud architectures, configuring identity and access management (IAM), monitoring cloud workloads, ensuring regulatory compliance, and remediating misconfigurations.

Technical Skills: Cloud platform administration (AWS, Azure, GCP), infrastructure-as-code security, container security, serverless security, encryption management.

Soft Skills: Adaptability, cross-functional collaboration, business acumen, continuous learning mindset.

Tools Used: AWS Security Hub, Azure Defender, Prisma Cloud, Terraform, CloudTrail, Prowler, ScoutSuite.

Certifications: CCSP (ISC²), AWS Certified Security – Specialty, AZ-500 (Microsoft), CCSK (CSA), Google Professional Cloud Security Engineer.

Career Path: Cloud Security Engineer → Senior Cloud Architect → Cloud Security Manager → VP of Cloud Infrastructure.

7. Network Security

Definition & Overview: Network security encompasses the policies, technologies, and practices designed to protect the integrity, confidentiality, and availability of computer networks. It involves defending against unauthorized access, misuse, and attacks at every layer of the OSI model.

Key Responsibilities: Designing and managing firewalls, configuring intrusion detection/prevention systems, segmenting networks, monitoring traffic for anomalies, and enforcing access control policies.

Technical Skills: TCP/IP protocols, firewall configuration, VPN management, IDS/IPS tuning, network segmentation, packet analysis.

Soft Skills: Problem-solving, analytical skills, documentation discipline, teamwork, time management.

Tools Used: Cisco ASA/Firepower, Palo Alto Networks, Snort, Suricata, Wireshark, pfSense, Zeek.

Certifications: CompTIA Security+, CCNA Security (Cisco), PCNSA (Palo Alto), GIAC GCIA, Fortinet NSE.

Career Path: Network Security Analyst → Network Security Engineer → Security Architect → Network Security Director.

8. Application Security

Definition & Overview: Application security (AppSec) focuses on identifying, fixing, and preventing security vulnerabilities within software applications throughout the development lifecycle. It integrates security into DevOps workflows, often referred to as DevSecOps.

Key Responsibilities: Performing code reviews and static/dynamic analysis, integrating security into CI/CD pipelines, conducting threat modeling for applications, and training developers on secure coding practices.

Technical Skills: Secure coding principles (OWASP Top 10), SAST/DAST/IAST tools, API security testing, container security scanning, programming proficiency.

Soft Skills: Collaboration with developers, communication, creative problem-solving, empathy, mentoring ability.

Tools Used: SonarQube, Checkmarx, Veracode, Snyk, OWASP ZAP, Semgrep, GitHub Advanced Security.

Certifications: CSSLP (ISC²), GWAPT (GIAC), CASE (EC-Council), OSWE (Offensive Security).

Career Path: Application Security Analyst → AppSec Engineer → Senior AppSec Architect → VP of Product Security.

9. Governance, Risk & Compliance (GRC)

Definition & Overview: GRC encompasses the frameworks, policies, and processes organizations use to align IT security with business objectives, manage risks, and ensure compliance with laws and regulations such as GDPR, HIPAA, PCI-DSS, and SOX.

Key Responsibilities: Developing security policies and procedures, conducting risk assessments, managing audit processes, ensuring regulatory compliance, and reporting risk posture to executive leadership.

Technical Skills: Risk assessment frameworks (NIST, ISO 27001), policy development, audit management, data classification, regulatory mapping.

Soft Skills: Strategic thinking, leadership, communication with non-technical stakeholders, negotiation, organizational skills.

Tools Used: RSA Archer, ServiceNow GRC, OneTrust, Qualys Policy Compliance, Vanta, Drata.

Certifications: CISM (ISACA), CISA (ISACA), CRISC (ISACA), CISSP (ISC²), ISO 27001 Lead Auditor.

Career Path: GRC Analyst → Compliance Manager → Risk Director → Chief Information Security Officer (CISO).

10. Red Team / Blue Team Operations

Definition & Overview: Red Team and Blue Team operations represent adversarial simulation and defense. The Red Team emulates sophisticated threat actors to test an organization's defenses, while the Blue Team detects, responds to, and mitigates those simulated attacks. Purple Team exercises combine both for continuous improvement.

Key Responsibilities: Red Team: Planning and executing full-scope attack simulations, social engineering campaigns, and physical security assessments. Blue Team: Detecting intrusions, hardening defenses, and improving response procedures based on findings.

Technical Skills: Red: Advanced exploitation, C2 frameworks, social engineering, evasion techniques. Blue: SIEM correlation, EDR analysis, threat hunting, forensic investigation.

Soft Skills: Adversarial mindset (Red), defensive creativity (Blue), collaboration, adaptability, clear reporting for both teams.

Tools Used: Red: Cobalt Strike, Covenant, Mythic, Bloodhound. Blue: Splunk, Velociraptor, YARA, Sigma rules, Elastic Security.

Certifications: Red: OSCP, CRT0 (Zero-Point Security), GXPN (GIAC). Blue: BTL1, CySA+, GCIA (GIAC). Both: CISSP.

Career Path: Red/Blue Team Analyst → Senior Operator → Team Lead → Director of Adversarial Operations / Defensive Security.

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